$$\left(\frac{10}{y}\right)' = -\frac{10}{y^2}$$

$$-\frac{x}{y^2}$$

$$f(x;y) = \ln \frac{x}{y}$$

$$\frac{\partial f}{\partial x} = \frac{y}{x} \cdot \frac{1}{y} = \frac{1}{x}$$

$$\frac{\partial f}{\partial y} = \frac{\cancel{y}}{\cancel{x}} \cdot \left(-\frac{\cancel{x}}{y^2}\right) = -\frac{1}{y}$$

$$\frac{1}{x} \cdot x + \left(-\frac{1}{y}\right) \cdot y = p \cdot f(x;y)$$

$$1 - 1 = p\, f(x;y)$$

$$0 = p\, f(x;y)$$

$$p = 0$$

$$\frac{\partial f}{\partial y} = \left(e^{\frac{y}{x}}\right)'_y = e^{\frac{y}{x}} \cdot \left(\frac{1}{x}\right)$$

$$\frac{\partial f}{\partial x} = \left(e^{\frac{y}{x}}\right)' = e^{\frac{y}{x}} \cdot \left(\frac{-y}{x^2}\right)$$

$$\underbrace{e^{\frac{y}{x}} \cdot \left(-\frac{y}{x}\right)} + \underbrace{e^{\frac{y}{x}} \cdot \frac{y}{x}} = 0$$

$$\frac{x^3+y^3}{(x+y)}$$

$$x^2 - xy + y^2$$

$$\frac{\partial f}{\partial x} = 2x - y$$

$$\frac{\partial f}{\partial y} = 2y - x$$

$$2x^2 - yx + 2y^2 - \cancel{y}yx =$$

$$2x^2 + 2y^2 - 2xy = 2(x^2 + y^2 - xy)$$

$$f(x, y, z) = \left(x^3 + y^3 + z^3\right)^{1/3}$$

$$kx \quad ky \quad kz =$$

$$\left(k^3x^3 + k^3y^3 + k^3z^3\right)^{1/3} =$$

$$\left(R^3\left(x^3+y^3+z^3\right)\right)^{1/3} =$$

$$R\left(x^3+y^3+z^3\right)^{1/3}$$

$$\frac{\partial f}{\partial x} = \left(\left(x^3+y^3+z^3\right)^{1/3}\right)' = \frac{1}{3}\left(x^3+y^3+z^3\right)^{1/3-1}\cdot 3x^2$$

$$= \frac{x^2}{\left(x^3+y^3+z^3\right)^{2/3}}$$

$$\frac{\partial f}{\partial y} = \frac{y^2}{\left(x^3+y^3+z^3\right)^{2/3}} \qquad \frac{\partial f}{\partial z} = \frac{z^2}{\left(x^3+y^3+z^3\right)^{2/3}}$$

$$\frac{x\cdot x^2+y y^2+z z^2}{\left(x^3+y^3+z^3\right)^{2/3}} = \frac{x^3+y^3+z^3}{\left(\left(x^3+\cancel{x}y^3+z^3\right)^2\right)^{1/3}} =$$

$$\sqrt[3]{x^3+y^3+z^3}$$

$$\frac{x^3+y^3+z^3}{\sqrt[3]{x^3+y^3+z^3}} = \left(x^3+y^3+z^3\right)^{2/3}$$

$$f(x;y;z)=\frac{x^2+y^2+z^2}{x+y+z}$$

$$\frac{\partial f}{\partial x}=\frac{2x(x+y+z)-x^2-y^2-z^2}{(x+y+z)^2}=$$

$$\frac{x^2+2xy+2xz-y^2-z^2}{(x+y+z)^2}$$

$$\frac{\partial f}{\partial y}=\frac{2y(x+y+z)-x^2-y^2-z^2}{(x+y+z)^2}$$

$$\frac{\partial f}{\partial z}=\frac{2z(x+y+z)-x^2-y^2-z^2}{(x+y+z)^2}$$

$$x^3+2x^2y+2x^2z-y^2x-z^2x+y^3+2y^2x+2y^2z-x^2y-z^2y$$

$$+z^3+2z^2y+2z^2x-x^2z-y^2z$$

$$x^3+y^3+z^3+x^2y+x^2z+y^2x+z^2x+y^2z+z^2y$$

$$\frac{x^3+y^3+z^3+x^2y+x^2z+y^2x+z^2x+y^2z+z^2y}{(x+y+z)^2}$$

$$(x^2+y^2+z^2)(x+y+z) = x^3+yx^2+zx^2+y^3+xy^2+zy^2+z^3+z^2x+z^2y$$

$$\frac{(x^2+y^2+z^2)\cancel{(x+y+z)}}{(x+y+z)^{\cancel{2}}} =$$

$$\frac{x^2+y^2+z^2}{x+y+z}$$

$$f(x,y,z) = \frac{x+y}{z} + \frac{z}{x+y}$$

$$\frac{\partial f}{\partial x} = \frac{1}{z} - \frac{z}{(x+y)^2} \qquad \frac{x+5}{3} = \frac{1}{3}(x+5) =$$

$$\frac{\partial f}{\partial y} = \frac{1}{z} - \frac{z}{(x+y)^2} \qquad \frac{1}{3}\cdot 1 = \frac{1}{3}$$

$$\frac{\partial f}{\partial z} = -\frac{(x+y)}{z^2} + \frac{1}{x+y}$$

$$\frac{x}{z} - \frac{zx}{(x+y)^2} + \frac{y}{z} - \frac{zy}{(x+y)^2} - \frac{z(x+y)}{z^2} + \frac{z}{x+y}$$

$$\cancel{\frac{x+y}{z}} - \cancel{\frac{z(x+y)}{(x+y)^2}} + \cancel{\frac{z}{x+y}}$$

$$-\cancel{\frac{x+y}{z}} = 0$$

$$f(x,y) = \ln(x^2+y^2) = \ln(k^2(x^2+y^2)) = k^n \cdot \ln(x^2+y^2)$$

$$\frac{\partial f}{\partial x} = \frac{1}{x^2+y^2+1} \cdot 2x$$

$$\frac{\partial f}{\partial y} = \frac{1}{x^2+y^2+1} \cdot 2y$$

$$\frac{2x^2}{x^2+y^2+1} + \frac{2y^2}{x^2+y^2+1} = \frac{2x^2+2y^2}{x^2+y^2+1}$$

$$f(kx;ky) = k^n f(x;y)$$

$$e^{x^2+y^2} \qquad = e^{k^2(x^2+y^2)}$$

$$\sqrt{4-9x^2} = t$$

$$\int \frac{dx}{\sqrt{4-9x^2}} = \int \frac{dx}{2\sqrt{1-\frac{9x^2}{4}}}$$

$$4-9x^2 = t$$

$$dt = -18x\,dx$$

$$\frac{dt}{-18x} = dx$$

$$\left(\frac{3x}{2}\right)^2 =$$

$$\frac{3x}{2} = t$$

$$\frac{3}{2}dx = dt$$

$$dx = \frac{2dt}{3}$$

$$\int \frac{dx}{\sqrt{1-x^2}} = \arcsin x$$

$$\int \frac{\frac{\cancel{2}}{3}dt}{\cancel{2}\sqrt{1-t^2}} = \frac{1}{3}\arcsin t + C$$

$$= \frac{1}{3}\arcsin\left(\frac{3x}{2}\right) + C$$

$$\int \frac{x^3}{\sqrt{1+x^2}}\,dx$$

$$\sqrt{1+x^2} = t$$

$$dt = \frac{\cancel{2}x}{\cancel{2}\sqrt{1+x^2}}\,dx$$

$$\frac{2\sqrt{1+x^2}}{2x}\,dt = dx$$

$$\int \frac{x^{\cancel{3}}}{\cancel{\sqrt{1+x^2}}}\,\frac{\cancel{\sqrt{1+x^2}}}{\cancel{x}}\,dt = \int x^2\,dt$$

$$1+x^2 = t^2$$
$$x^2 = t^2-1$$

$$\int (t^2 - 1)\,dt = \frac{t^3}{3} - t + C$$

$$\frac{(1+x^2)^{3/2}}{3} - \sqrt{1+x^2} + C$$